

GO / NO GO: Intelligent Sea Breeze Prediction System for Water Sports Using Machine Learning

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Abstract

This study investigates whether Sea Breeze events in Badalona (Spain) can be predicted using historical atmospheric and oceanographic data. The objective was to develop a machine learning model capable of generating GO/NO GO recommendations for water sports enthusiasts. Three real-world datasets covering the period 2010–2025 were integrated: local wind observations, meteorological measurements, and oceanographic buoy records. More than four million raw observations were processed and aggregated into 5,806 daily records. Feature engineering focused on physical mechanisms associated with sea breeze formation, including solar irradiance, land–sea thermal contrast, night–day temperature gradient, sea surface temperature, and atmospheric pressure. A Random Forest classifier was trained using a strict temporal split strategy to avoid information leakage. Results show that the model achieved an F1-score of 0.518 and a ROC-AUC of 0.714 on the test dataset, demonstrating meaningful predictive capability. The findings confirm that sea breeze occurrence is driven by identifiable atmospheric patterns and can be predicted with useful accuracy for operational decision support.

Introduction

Sea Breeze Wind is a key factor for windsurfing, kitesurfing, sailing and other water sports along the Mediterranean coast. Traditional weather forecasts often fail to capture local thermal effects, creating uncertainty for users who must decide whether to travel to a wind spot. This project addresses the question: Can machine learning predict the occurrence of favorable Sea Breeze conditions in advance? The study aims to identify the atmospheric drivers of thermal wind formation and evaluate the feasibility of a data-driven GO/NO GO recommendation system.

Methodology

The research followed a complete data analytics pipeline. Three datasets were integrated: (1) Buscaviento Badalona wind observations (15-minute resolution), (2) XEMA/Meteocat meteorological records (hourly resolution), and (3) Barcelona oceanographic buoy data including Sea Surface Temperature (hourly resolution). Data covered the period 2010–2025. After cleaning, harmonization and temporal aggregation, a daily modeling dataset was generated. Feature engineering produced physically meaningful variables such as maximum solar irradiance, maximum air–sea thermal gradient, night–day temperature gradient, mean sea surface temperature, atmospheric pressure, precipitation, season, month and day of year. A Random Forest classifier with 500 trees, maximum depth of 12 and class balancing was selected due to its robustness and ability to model non-linear relationships. To simulate real operational deployment, a temporal split strategy was used: training (2010–2021), testing (2022–2023) and validation (2024–2025).

Results

Exploratory analysis revealed clear seasonal behavior, with Sea Breeze events occurring more frequently during warmer months. The probability of thermal wind increased as land–sea thermal contrast and night–day temperature differences increased. The Random Forest model achieved the following performance on the test dataset: Accuracy = 0.648, Precision = 0.447, Recall = 0.616, F1-score = 0.518 and ROC-AUC = 0.714. Validation performance remained stable (F1 = 0.470; ROC-AUC = 0.701), indicating limited overfitting. Feature importance analysis identified maximum solar irradiance (23.1%), air–sea thermal gradient (16.5%), night–day gradient (15.0%), day of year (13.6%), mean atmospheric pressure (12.0%) and sea surface temperature (11.7%) as the most influential predictors.

Discussion

The results support the hypothesis that Sea Breeze formation is not random but driven by measurable physical mechanisms. The prominence of solar irradiance and thermal gradients aligns with established meteorological theory describing thermal circulation between land and sea. The model successfully captured these relationships without using observed wind variables, reducing the risk of data leakage and improving scientific validity. Although performance is not sufficient for fully autonomous operational forecasting, the model clearly outperforms random or majority-class approaches. From a practical perspective, the recall value of 0.616 is particularly relevant because missing a strong thermal wind event is often more costly for users than generating a false positive recommendation. The study also demonstrates the value of combining meteorological and oceanographic information. Future improvements could include numerical weather prediction outputs, early-morning predictors and real-time forecast data to transform the system into a genuine operational forecasting platform.

Conclusion

This project demonstrates that machine learning can successfully identify atmospheric conditions associated with Sea Breeze events in Badalona. By integrating sixteen years of meteorological, oceanographic and wind observations, the study confirmed the importance of solar radiation and thermal gradients in sea breeze development. The Random Forest model achieved a ROC-AUC of 0.714 and showed stable temporal generalization. These findings provide a solid foundation for the future development of an operational GO/NO GO decision-support system for water sports users.

References (APA)

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